Machine Learning

17-04-26

KMeans Clustering

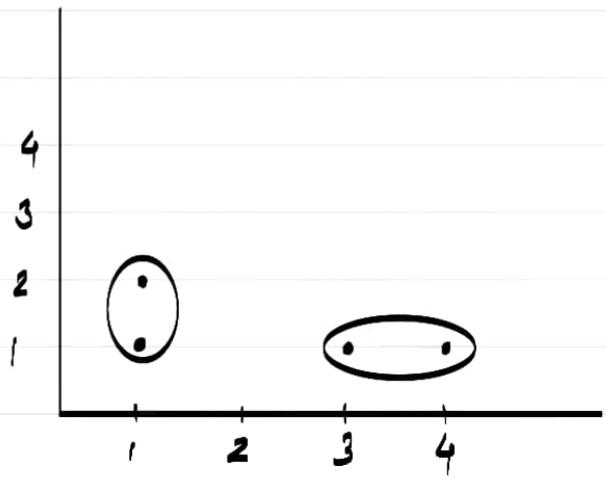
Training

Step 1 - decide value of K = no. of clusters
lets say we

Step 2 - randomly assign K centroids for K clusters

here we need 2 centroids

	x_1	x_2
d_1	1	1
d_2	1	2
d_3	3	2
d_4	4	2



randomly pick 2 datapoints and assign them as centroids

$C_1 = d_1 = (1, 1)$

$C_2 = d_2 = (1, 2)$

Step 3 - assign all train datapoints to their nearest centroid

d_1	d_2	d_3	d_4
$(1,1)$	$(1,2)$	$(3,2)$	$(4,2)$

$C_1 = (1,1)$ $\qquad$ $C_2 = (1,2)$

	dist with C_1	dist with C_2	assigned centroid
d_1	0	1	C_1
d_2	1	0	C_2
d_3	$\sqrt{5}$	$\sqrt{4}$	C_2
d_4	$\sqrt{10}$	$\sqrt{9}$	C_2

Step 4 - update the centroids using new datapoints which are assigned to them

update centroid C_1

current $C_1 = (1,1)$

assigned datapts = $d_1 = (1,1)$

find new centroid as average of assigned datapoints

new $C_1 = \left[\frac{1}{1}, \frac{1}{1}\right] = (1,1)$

update centroid C_2

current $C_2 = (1,2)$

assigned datapts: d_2 $(1,2)$ $\quad$ d_3 $(3,2)$ $\quad$ d_4 $(4,2)$

new $C_2 = \left[\frac{1+3+4}{3}, \frac{2+2+2}{3}\right] = \left(\frac{8}{3}, 2\right)$

$c_1 = (1,1)$ $\qquad$ $c_2 = (8/3, 2)$

Step 5- repeat step3 and step4 untill all centroids stop getting updated

$c_1 = (1,1)$ $\qquad$ $c_2 = (8/3, 2)$

$dist(d_1\ c_1) = 0$

$dist(d_2\ c_1) = [(1-1)^2 + (2-1)^2]^{1/2} = 1$

$dist(d_2\ c_2) = \sqrt{(1-\frac{8}{3})^2 + (2-2)^2} = \left[\left(\frac{3-8}{3}\right)^2 + 0\right]^{1/2} = \frac{5}{3}$

assign c_1 for d_2 as $dist(d_2\ c_1) < dist(d_2\ c_2)$

$dist(d_3\ c_1) = [(1-3)^2 + (1-2)^2]^{1/2} = 5^{1/2}$

$dist(d_3\ c_2) = [(3-8/3)^2 + (2-2)^2]^{1/2} = [\frac{1}{9} + 0]^{1/2} = \left(\frac{1}{9}\right)^{1/2}$

assign c_2 for d_3 as $dist(d_3\ c_2) < dist(d_3\ c_1)$

$dist(d_4\ c_1) = [(4-1)^2 + (2-1)^2]^{1/2} = \sqrt{10}$

$dist(d_4\ c_2) = [(4-8/3)^2 + (2-2)^2]^{1/2} = [(\frac{4}{3})^2 + 0]^{1/2} = \sqrt{\frac{16}{9}}$

assign c_2 for d_4 as $dist(d_4\ c_2) < dist(d_4\ c_1)$

$$\left.\begin{array}{l} c_1 \rightarrow d_1\ d_2 \\ c_2 \rightarrow d_3\ d_4 \end{array}\right\} \text{update centroids}$$

new c_1 = avg of d_1 d_2

$$c_1 = \left[\frac{(1+1)}{2}, \frac{(1+2)}{2} \right] = (1, 3/2)$$

new c_2 = avg of d_3 d_4

$$c_2 = \left[\frac{(3+4)}{2}, \frac{(2+2)}{2} \right] = (7/2, 2)$$

$c_1 = (1, 3/2)$ $\qquad$ $c_2 = (7/2, 2)$

again repeat step 3 and 4

after step 3

$c_1 \rightarrow d_1 \quad d_2$

$c_2 \rightarrow d_3 \quad d_4$

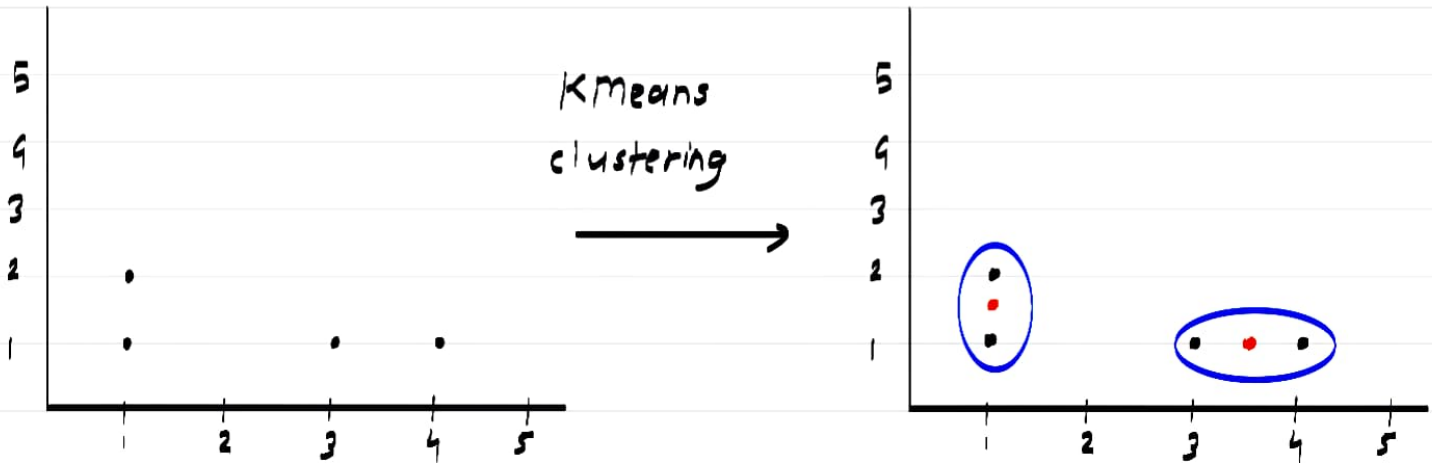
after step 4

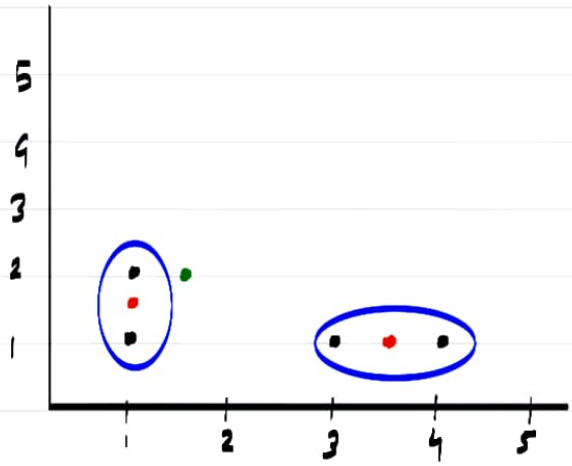
$c_1 = (1, 3/2)$
$c_2 = (7/2, 2)$

} updated centroids will be exactly same as the previous ones

↓

this means centroid updation has stopped





$C_1 = (1, 3/2)$

$C_2 = (7/2, 2)$

new datapoint $d_5 = (1.5, 2)$

$$dist(d_5, C_1) = [(1.5-1)^2 + (2-1.5)^2]^{1/2} = [0.25 + 0.25]^{1/2} = \sqrt{0.5}$$

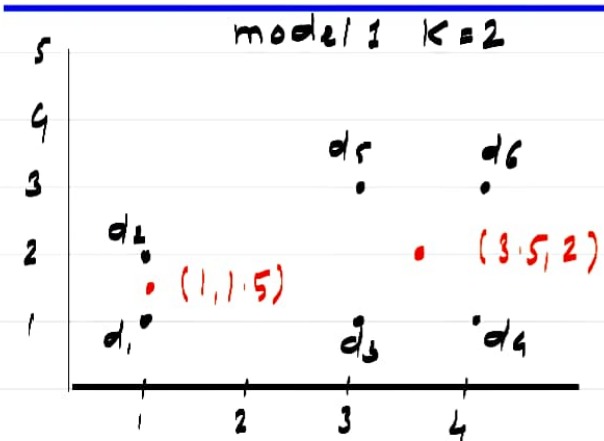
$$dist(d_5, C_2) = [(1-3.5)^2 + (2-2)^2]^{1/2} = \sqrt{4}$$

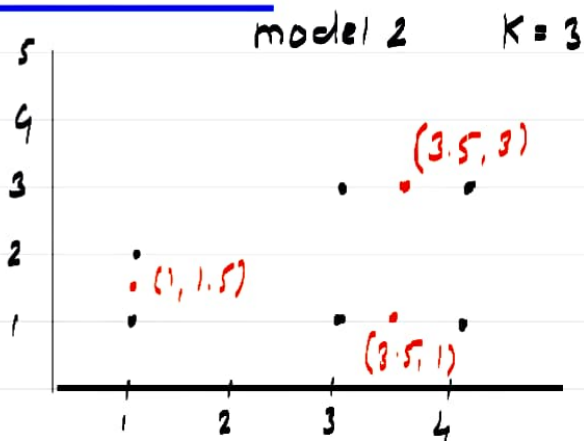
d_5 belongs to cluster C_1 as
$dist(d_5, C_1) < dist(d_5, C_2)$

How to find best value of K?
K is a hyperparameter here

Evaluation of clustering model

Within clusters sum of squares (WCSS)





WCSS computation

model 1

$c_1 = (1, 1.5)$ $\quad c_1 \rightarrow d_1, d_2$

$dist(d_1, c_1) = [(1-1)^2 + (1-1.5)^2]^{1/2} = 0.5$

$dist(d_2, c_1) = [(1-1)^2 + (2-1.5)^2]^{1/2} = 0.5$

$c_2 = (3.5, 2)$ $\quad c_2 \rightarrow$ d_3 $(3,1)$, d_4 $(4,1)$, d_5 $(3,3)$, d_6 $(4,3)$

$dist(d_3, c_2) = [(3-3.5)^2 + (2-2)^2]^{1/2} = 0.5$

$dist(d_4, c_2) = [(4-3.5)^2 + (2-2)^2]^{1/2} = 0.5$

$dist(d_5, c_2) = [(3-3.5)^2 + (3-2)^2]^{1/2} = \sqrt{1.25}$

$dist(d_6, c_2) = [(4-3.5)^2 + (3-2)^2]^{1/2} = \sqrt{1.25}$

WCSS: sum of squares of distances betn datapoints and their respective centroid

$$WCSS = \sum_{i=1}^{6} \sum_{j=1}^{2} [dist(d_i, c_j)]^2$$

$WCSS = 0.5^2 + 0.5^2 + 0.5^2 + 0.5^2 + 1.25 + 1.25$

$WCSS = 3.5 \longrightarrow$ model 1

lesser the WCSS $\Longrightarrow$ better the clustering

performance/accuracy of clustering $\propto \dfrac{1}{WCSS}$

if wcss model 2 < wcss model 1 then

model 2 is <u>better</u>.